University of Mumbai
Examination June 2021

Examinations Commencing from 1st June 2021

Program: 1T00728 / / B.E. (Computer Engineering)(SEM-VIII)
(Choice Base Credit Grading System)(R2016)
Curriculum Scheme: Rev 2016
Examination: BE Semester VIII
Course Code: CSC802 and Course Name: Distributed Computing

Time: 2 hour Max. Marks: 80

===

Q1.	**Choose the correct option for following questions. All the Questions are compulsory and carry equal marks. (2 marks each)**
1.	A layer which lies between an operating system and the applications running on it is called as -
Option A:	Firmware
Option B:	Hardware
Option C:	Software
Option D:	Middleware
2.	Goals of Distributed system does not include-
Option A:	Resource sharing
Option B:	Access to remote resources
Option C:	Sharing memory space
Option D:	Concurrent process execution
3.	which of the following is not the commonly used semantics for ordered delivery of multicast messages-
Option A:	Absolute ordering
Option B:	Persistent ordering
Option C:	Consistent ordering
Option D:	Casual ordering
4.	The type of transparency that enables resources to be moved while in use without being noticed by users and application is-
Option A:	Location Transparency
Option B:	Migration Transparency
Option C:	Relocation Transparency
Option D:	Access Transparency
5.	A paradigm of multiple autonomous computers, having a private memory, communicating through a computer network, is known as-
Option A:	Distributed computing
Option B:	Cloud computing
Option C:	Centralized computing
Option D:	Parallel computing

6.	Following is not the common mode of communication in Distributed system-
Option A:	RPC
Option B:	RMI
Option C:	Message Passing
Option D:	Shared memory
7.	Following is not the physical clock synchronization algorithm-
Option A:	Lamport's Scalar Clock synchronization
Option B:	Christians clock synchronization
Option C:	Berkley clock synchronization
Option D:	Network time protocol
8.	Distributed Mutual Exclusion Algorithm does not use-
Option A:	Coordinator process
Option B:	Token
Option C:	Logical clock for event ordering
Option D:	Request and Reply message
9.	Vector Timestamp Ordering Algorithm is an example of-
Option A:	Centralized Mutual Exclusion
Option B:	Distributed Mutual Exclusion
Option C:	Physical Clock Synchronization
Option D:	Logical Clock Synchronization
10.	What is fault tolerance in distributed Computing?
Option A:	Ability of system to continue functioning in the event of a complete failure.
Option B:	Ability of system to continue functioning in the event of a partial failure.
Option C:	Ability of system to continue functioning when system is properly working.
Option D:	Ability of distributed system to work in all conditions.
11.	In Task Assignment Approach, we have to-
Option A:	Minimize IPC cost
Option B:	Maximize IPC cost
Option C:	Fix IPC cost
Option D:	Keep constant IPC cost
12.	Backward error recovery requires-
Option A:	Grouping
Option B:	Assurance
Option C:	Check pointing
Option D:	Validation
13.	Which of these consistency models does not use synchronization operations?
Option A:	Sequential
Option B:	Weak
Option C:	Release
Option D:	Entry
14.	Which is not possible in distributed file system?

Option A:	File replication
Option B:	Migration
Option C:	Client interface
Option D:	Remote access
15.	X.500 is a-
Option A:	Directory services
Option B:	Naming services
Option C:	Replication services
Option D:	Consistency services
16.	A DFS is executed as a part of-
Option A:	System specific program
Option B:	Operating system
Option C:	File system
Option D:	Application program
17.	Processes on the remote systems are identified by-
Option A:	Host ID
Option B:	Identifier
Option C:	Host name and identifier
Option D:	Process ID
18.	The function of load-balancing algorithm is-
Option A:	It tries to balance the total system load by transparently transferring the workload from heavily loaded nodes to lightly loaded
Option B:	It helps the process to know the time by simply making a call to the operating system.
Option C:	allows a process to access named entity
Option D:	It synchronizes the clocks
19.	A Multi-threaded Server has following threads-
Option A:	Dispatcher Thread
Option B:	Client Thread
Option C:	Worker Thread
Option D:	Client and Server Thread
20.	Maekawa's Mutual Exclusion Algorithm is based on-
Option A:	Coordinator selection
Option B:	Token
Option C:	Voting
Option D:	Tickets

Q2 **(20 Marks)**	***Solve the following questions.***	
A	**Solve any Two**	**5 marks each**
i.	Write short note on - Group Communication.	

ii.	What is replication in distributed system? Explain the advantages of replication.
iii.	Write short note on - Network File System (NFS)
B	**Solve any One** **10 marks each**
i.	Explain the Centralized algorithms for Mutual Exclusion in Distributed Systems.
ii.	Describe File caching schemes in brief.

Q3. **(20 Marks)**	***Solve the following questions.***
A	**Solve any Two** **5 marks each**
i.	Discuss the Bully algorithm with appropriate example. State its advantages and disadvantages.
ii.	What are the different model of distributed system? Explain.
iii.	How Monotonic Read consistency model is different from Read your Write consistency Model? Support your answer with suitable example.
B	**Solve any One** **10 marks each**
i	What is the need for Code Migration? Explain the code migration issues in detail.
ii	Define remote procedure call (RPC)? Describe the working of RPC in detail.